Math 81: Abstract Algebra

Prishita Dharampal

Credit Statement: Talked to Sair Shaikh'26, and Math Stack Exchange.

Problem 1. Let F be a field and K/F and L/F be subextensions of a field extension M/F.

(a) Prove that if $K = F(\alpha_1, \dots, \alpha_n)$ and $L = F(\beta_1, \dots, \beta_m)$ are finitely generated, then

$$KL = F(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m).$$

(b) Prove that if K/F and L/F are finite then KL/F is finite and

$$[KL : F] \leq [K : F][L : F],$$

with equality if and only if K/F and L/F are linearly disjoint.

Hint. Prove that if $x_1, \dots, x_n$ is an F-basis for K and $y_1, \dots, y_m$ is an F-basis for L, then the products $x_i y_j$ for $1 \leq i \leq n$ and $1 \leq j \leq m$ span KL/F and are an F-basis if and only if K/F and L/F are linearly disjoint.

(c) Prove that finite extensions K/F and L/F of relatively prime degree are linearly disjoint.

(d) Prove that if K/F and L/F are linearly disjoint then $K \cap L = F$. Find an example showing that the converse is false.

(e) Prove that if K/F and L/F are linearly disjoint finite Galois extensions then KL/F is Galois and the map

$$\mathrm{Gal}(KL/F) \to \mathrm{Gal}(K/F) \times \mathrm{Gal}(L/F), \quad \sigma \mapsto (\sigma|_K, \sigma|_L)$$

is a group isomorphism.

Solution.

(a) $(\Leftarrow)$ $KL \supseteq F(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m)$

Note that KL is defined as the smallest subfield of M containing both L and K. That is,

$$F(\alpha_1, \ldots, \alpha_n) \subset KL \text{ and } F(\beta_1, \ldots, \beta_m) \subseteq KL$$

So KL contains each α_i and each β_j, and hence $F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m) \subseteq KL$.

($\Rightarrow$) $KL \subset F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m)$
Let $E = F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m)$ Note that

$$K = F(\alpha_1, \ldots, \alpha_n) \subset E \text{ and } L = F(\beta_1, \ldots, \beta_m) \subset E$$

Since KL is the smallest fields containing K and L, and E contains both K and L,

$$KL \subseteq E = F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m)$$

That is $KL = F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m)$.

(b) Let $K = (\alpha_1, \ldots, \alpha_n)$, and $L = (\beta_1, \ldots, \beta_n)$. Also note, from part (1), $KL = F(\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m)$, or equivalently $KL = L(\alpha_1, \ldots, \alpha_n)$.

Now, we get the following towers:

$$\begin{array}{ccc}
K = K_{n-1}(\alpha_n) & \qquad & KL = KL_{n-1}(\alpha_n) \\
\uparrow n_n & & \uparrow m_n \\
K_{n-1} = K_{n-2}(\alpha_{n-1}) & & KL_{n-1} = KL_{n-2}(\alpha_{n-1}) \\
\uparrow n_{n-1} & & \uparrow m_{n-1} \\
\vdots & & \vdots \\
\uparrow n_2 & & \uparrow m_2 \\
K_1 = F(\alpha_1) & & KL_1 = L(\alpha_1) \\
\uparrow n_1 & & \uparrow m_1 \\
F & & L
\end{array}$$

where,

$$[K : F] = [K : K_{n-1}] \cdot [K_{n-1} : K_{n-2}] \cdot \ldots \cdot [K_1 : F]$$

and,

$$[KL : L] = [KL : KL_{n-1}] \cdot [KL_{n-1} : KL_{n-2}] \cdot \ldots \cdot [KL_1 : L]$$

Note that at each step,

$$[KL_i : KL_{i-1}] \leq [K_i : K_{i-1}]$$

because the minimal polynomial of α_i over KL_{i-1} divides the minimal polynomial of α_i over K_{i-1}, i.e. the degree can only decrease. Combining these inequalities (over n steps) gives us,

$$[KL : L] \leq [K : F]$$

By Tower Law,

$$[KL : F] = [KL : L][L : F]$$

Substituting in the inequality,

$$[KL : F] \leq [K : F][L : F]$$

Hence proved.

Now we show that $[KL : F] \leq [L : F][K : F]$ if and only if K and L are linearly disjoint. Let the F-basis for K as a vector space be $\{1 = \alpha_1, \alpha_2, \ldots, \alpha_n\}$ and the F-basis for L as a vector space be $\{1 = \beta_1, \beta_2, \ldots, \beta_m\}$. Also, let $[K : F] = n, [L : F] = m$.

$(\Longleftarrow)$

Assume K and L are linearly disjoint.

Then in K/F, we know that if

$$a_1\alpha_1 + \ldots + a_n\alpha_n = 0 \qquad a_i \in F$$

then $a_i = 0$, for all i. Since K and L are linearity disjoint, any F-linearly independent subset of K is L-linearly independent in KL, that is, if

$$\lambda_1\alpha_1 + \ldots + \lambda_n\alpha_n = 0 \qquad \lambda_i \in L$$

then $\lambda_i = 0$, for all i.

As $\lambda_i \in L$ we can write it as a linear combination such that $\lambda_i = \sum_{j=1}^{k} a_j\beta_j$, where $a_j \in F$ and β_j is in the basis, and m is the size of the basis. Substituting,

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij}\beta_j\alpha_i = 0$$

Then, $a_{ij} = 0$, for all ij. By linear disjoint-ness $\lambda_i = 0$, i.e. $sum_{j=1}^{m} a_{ij}\beta_j = 0$. But by linear indepence of β_is, $a_{ij} = 0$, and hence $\{\alpha_i\beta_j\}$ are linearly independent for all i, j. Since $[KL : F] \leq n \cdot m$ we have $n \cdot m$ linearly independent elements, this is a maximal linearly independent set and hence a basis for KL/F. That is, $[KL : F] = nm = [K : F][L : F]$.

($\implies$)

Assume $[KL : F] = [K : F][L : F] = nm$.

We want to show that the F-basis of K is L-linearly independent in KL. Consider the set,

$$\{\alpha_i\beta_j : 1 \le i \le n, 1 \le j \le m\}$$

Any element of KL is an F-linear combination of products of elements from K and L, i.e. the set $\{\alpha_i\beta_j\}$ spans KL over F. The set has nm spanning elements, and since $[KL : F] = nm$, they must in fact form a F-basis.

Now suppose $\sum_{i=1}^{n} \lambda_i\alpha_i = 0$ for some $\lambda_i \in L$. Writing $\lambda_i = \sum_{j=1}^{m} a_{ij}\beta_j$ with $a_{ij} \in F$, we get:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij}\alpha_i\beta_j = 0$$

Since $\{\alpha_i\beta_j\}$ is an F-basis, $a_{ij} = 0$ for all i, j, and hence $\lambda_i = 0$ for all i. Thus K and L are linearly disjoint.

(c)

$$\begin{tikzcd}
& KL & \\
K \arrow[ur, "a"] & & L \arrow[ul, "b"'] \\
& F \arrow[ul, "n"] \arrow[ur, "m"'] \arrow[uu, "k"] &
\end{tikzcd}$$

Note that by tower law $a \cdot n = m \cdot b$ and $an|k$, $bm|k$. If $\gcd(n, m) = 1$, $k \ge nm$. Since $k \ge nm$ and $k \le nm$ (part (2)), $k = nm$. And from part (2), we know that $[KL : F] = nm$ if and only if K and L are linearly disjoint. Thus, K and L are disjoint.

(d) Assume $K = F(\alpha_2, \dots, \alpha_n)$ and $L = F(\beta_2, \dots, \beta_n)$ are linearly disjoint. Let the basis for K as a vector space be $\{1, \alpha_2, \dots, \alpha_n\}$ and the basis for L as a vector space be $\{1, \beta_2, \dots, \beta_n\}$.

Since both K and L are field extensions of F, the fact that $F \subseteq K \cap L$ is obvious.

Now, consider $x \in K \cap L$. Since $x \in K$, we can write,

$$x = a_1 \cdot 1 + \sum_{i=2}^{n} a_i\alpha_i \quad a_1, a_i \in F$$

and, since $x \in L$,

$$x = -b_1 \cdot 1 + \sum_{j=2}^{m} -b_j\beta_j \quad b_1, b_j \in F$$

Then,

$$a_1 \cdot 1 + \sum_{i=2}^{n} a_i\alpha_i + b_1 \cdot 1 + \sum_{i=2}^{m} b_j\beta_j = 0$$
$$(a_1 + b_1) \cdot 1 + \sum_{i=2}^{n} a_i\alpha_i + \sum_{j=2}^{m} b_j\beta_j = 0$$

We know that $1, \alpha_i, \beta_j$ are linearly independent (part (2)). Then, for all $i \in \{2, \dots, n\}$ and $j \in \{2, \dots, m\}$

$$a_1 + b_1 = a_i = b_j = 0$$

That implies,

$$\sum_{i=2}^{n} -a_i\alpha_i = 0 \text{ and } x = a_1 \cdot 1$$

And,

$$\sum_{j=2}^{m} -b_j\beta_j = 0 \text{ and } x = b_1 \cdot 1$$

But $a_1, b_1 \in F$, i.e. $x \in F$. Since x was an arbitary element in $K \cap L$, this proves the opposite containment.

$$K \cap L \subseteq F$$

Hence,

$$K \cap L = F$$

CounterExample:

Let $F = \mathbb{Q}, K = \mathbb{Q}(\sqrt[3]{2}), L = \mathbb{Q}(\omega\sqrt[3]{2})$. Then, $K \cap L = F = \mathbb{Q}$. But $[K : F] = 3, [L : F] = 3$ but $[KL : F] = 6 \neq 9$. Hence they are not linearly disjoint.

(e) Assume K/F and L/F are linearly disjoint finite Galois extensions.

Since K/F is Galois, K is the splitting field of some irreducible separable $f(x) \in F[x]$. Similarly, L is the splitting field of some irreducible seperable $g(x) \in F[x]$. Also, since K and L are linearly disjoint (part (d)), $K \cap L = F$, i.e. f, g don't share roots.

Now, consider the seperable polynomial $fg(x) \in F[x]$. Let it's splitting field be R. We want to show that $R = KL$.

The containment $R \subseteq KL$ is obvious, since the respective splitting fields of f and g are subfields of KL.

Note that $f(x)$, and $g(x)$ split over R. By minimality of splitting fields, $K \subset R$ and $L \subset R$. But KL is the smallest field containing both K and L, i.e. $KL \subset R$.

Hence, $R = KL$. Since KL is the splitting field of a separable irreducible polynomial over $F[x]$, it is Galois.

Now consider the map

$$\begin{aligned} \phi : \mathrm{Gal}(KL/F) &\to \mathrm{Gal}(K/F) \times \mathrm{Gal}(L/F) \\ \sigma &\mapsto (\sigma|_K, \sigma|_L) \end{aligned}$$

Let $[K : F] = |\mathrm{Gal}(K/F)| = n, [L : F] = |\mathrm{Gal}(L/F)| = m$, that is $\mathrm{Gal}(K/F) \times \mathrm{Gal}(L/F) = nm$. Also note that since K and L are linearly disjoint, part (b) gives us

$$[KL : F] = nm = \mathrm{Gal}(KL/F)$$

Note that ϕ is a homomorphism because for any $\sigma_a, \sigma_b \in \mathrm{Gal}(KL/F)$,

$$\begin{aligned} \phi(\sigma_a \circ \sigma_b) &= ((\sigma_a \circ \sigma_b)|_K,\ (\sigma_a \circ \sigma_b)|_L) \\ &= (\sigma_a|_K \circ \sigma_b|_K,\ \sigma_a|_L \circ \sigma_b|_L) \\ &= (\sigma_a|_K, \sigma_a|_L) \cdot (\sigma_b|_K, \sigma_b|_L) \quad \text{(componentwise multiplication)} \\ &= \phi(\sigma_a) \cdot \phi(\sigma_b) \end{aligned}$$

Let $\sigma \in \ker(\phi)$. Then,

$$\phi(\sigma) = (\sigma|_K,\ \sigma|_L) = (id_K, id_L)$$

That is, $\sigma|_K = \mathrm{id}_K$ and $\sigma|_L = \mathrm{id}_L$, meaning σ fixes every element of K and every element of L. Since every element of KL is a product of elements from K and L (part (b)), σ fixes all of KL, so $\sigma = \mathrm{id}_{KL}$ and $\ker\phi$ is trivial. An injective homomorphism between groups of the same size is surjective, and hence ϕ is a group isomorphism.

Problem 2. Let K/F be a finite extension of fields and $f(x) \in F[x]$ an irreducible polynomial of degree n.

(a) Prove that if n does not divide $[K : F]$ then $f(x)$ has no roots in K.

(b) Prove that if n is relatively prime to $[K : F]$ then $f(x)$ is irreducible over K.

Solution.

1. Let α be a root to a minimal polynomial $f(x)$ then the extension

$$[F(\alpha) : F] = \deg(m_{\alpha/F}(x)) = n$$

If some root $\alpha \in K$ then $F(\alpha) \subset K$. Then by tower law,

$$[K : F(\alpha)][F(\alpha) : F] = [K : F] \implies [F(\alpha) : F]\Big|[K : F]$$

But we know that $n \nmid [K : F]$, that is there are no elements of degree n over F in K and hence no roots of $f(x) \in K$

2. Assume that $\gcd(n, [K : F]) = 1$, and $f(x)$ is irreducible over F.

We will prove this by contradiction.

Assume $f(x)$ is not irreducible over K. Then it splits into some irreducible polynomials $g_i(x) \in K[x]$ such that $f(x) = \prod g_i(x)$. Consider one such $g(x)$ over K. Let the degree of this polynomial be d, since $g(x)$ is irreducible over K it is the minimal polynomial for some root α. Then,

$$[K(\alpha) : K] = d$$

And since $f(x)$ is irreducible over F,

$$[F(\alpha) : F] = n$$

Since $F \subseteq F(\alpha) \subseteq K(\alpha)$ and $F \subseteq K \subseteq K(\alpha)$, applying the Tower Law twice gives:

$$[K(\alpha) : F] = [K(\alpha) : K][K : F] = d \cdot [K : F]$$
$$[K(\alpha) : F] = [K(\alpha) : F(\alpha)][F(\alpha) : F] = [K(\alpha) : F(\alpha)] \cdot n$$

Hence $n \mid d \cdot [K : F]$. Since $\gcd(n, [K : F]) = 1$,

$$n \mid d$$

But since $g(x) \mid f(x)$, we have $d \leq n$, and so $d = n$. Hence $f(x) = g(x)$ up to scalar multiples, i.e. $f(x)$ is irreducible over K.

Problem 3. Let F be a field, $f(x)$ a polynomial of degree n over F with splitting field E/F.

(a) Let K/F be a subextension of E/F. Prove that E/K is a splitting field of $f(x)$ considered as a polynomial over K.

(b) Prove that $[\mathsf{E} : \mathsf{F}]$ divides $n!$.

Hint. Use induction on n, and deal with the cases of f reducible or irreducible separately. At some point you will need the fact that $a!b!$ divides $(a+b)!$, which you should also prove (or have already proved in Math 71).

Solution.

1. Let $F \subset K \subset E$, where $E = F(\alpha_1, \ldots, \alpha_n)$ where $\alpha_1, \ldots, \alpha_n$ are the n roots of $f(x)$, and let E' be the splitting field of $f(x) \in K[x]$

   Consider $f(x) \in K[x]$. Since $F \subset K$, and $\alpha_1, \ldots, \alpha_n \in E$, $f(x)$ considered as a polynomial over K definitely splits over E. That is,

   $$E' \subseteq E$$

   Also, by definition, $E' = K(\alpha_1, \ldots, \alpha_n)$. Since $F \subset K, F(\alpha_1, \ldots, \alpha_n) \subset K(\alpha_1, \ldots, \alpha_n)$. Then, by minimality of splitting fields we can seee $E \subseteq E'$.

   Together, these inclusions give, $E' = E$. Hence, E/K is the splitting field of $f(x)$ as a polynomial over K.

2. We will prove that $[E : F]$ divides n by induction. Let $[E : F] = m$.

   Base Case: $n = 1$.

   Consider a linear polynomial $f(x)$ over F. Its splitting field $E = F$,

   $$f(x) = x - \alpha \implies f(\alpha) = 0$$

   That is, $[E : F] = 1$ and $1|1!$. Hence base case holds.

   Inductive Hypothesis: Assume that the degree of the splitting field of polynomials with degree $x < n$ divides $x!$.

   Inductive Step: For a polynomial $f(x) \in F[x]$ of degree n with splitting field E, there are two scenarios we must consider.

   (a) $f(x)$ is irreducible over F.
   Let α be a root of $f(x)$. So,

   $$[F(\alpha) : F] = n$$

In $F(\alpha)$, we have $f(x) = (x-\alpha)g(x)$, where $\deg(g(x)) = n-1$. By tower law,

$$[E : F] = [E : F(\alpha)][F(\alpha) : F]$$

The splitting field of $g(x)$ over $F(\alpha)$ is E, and by IH

$$[E : F(\alpha)] \Big| (n-1)!$$

Note that $(n-1)!n \Big| n!$.

(b) $f(x)$ is not irreducible over F.

Then $f(x) = g(x)h(x)$ for some $g(x), h(x) \in F[x]$. Note that both $g(x)$ and $h(x)$ have degree less than n. Let $\deg(g(x)) = a, \deg(h(x)) = b$, where $a+b=n$. Let E_g be the splitting field of g, then E is the splitting field of $h(x)$ over E_g. By tower law,

$$[E : F] = [E : E_g][E_g : F]$$

By inducton hypothesis, $[E_g : F] \Big| a!$ and $[E : E_g] \Big| b!$. We need to show that $a!b! \Big| n! = (a+b)!$.

Recall from Group Theory, that $|S_a| = a!$, $|S_b| = b!$, and $|S_n| = n! = (a+b)!$. Moreover, note that $S_a \times S_b \hookrightarrow S_{a+b}$ is an injective group homomorphism, so $S_a \times S_b$ is isomorphic to a subgroup of S_{a+b}. By Lagrange's Theorem:

$$|S_a \times S_b| \Big| |S_{a+b}|$$

$$a!\, b! \mid (a+b)!$$

Hence, the hypothesis holds.

Problem 4. Let K/F be a Galois extension with Galois group isomorphic to $C_2 \times C_{12}$. How many subextensions $K/M/F$ are there satisfying:

(a) $[K:M] = 6$

(b) $[M:F] = 6$

(c) $\mathsf{Gal}(K/M) \cong C_6$

(d) $\mathsf{Gal}(M/F) \cong C_6$

Solution.

We know from FTGT that for extensions of the form $K/M/F$, and the subgroup H of the Galois Group $G \cong C_2 \times C_{12}$ that corresponds to the fixed field $M = K^H$ the following hold,

1. $[K:M] = |H|$
2. $[M:F] = [G:H]$
3. $\mathsf{Gal}(K/M) \cong H$
4. $\mathsf{Gal}(M/F) \cong G/H$

Also note that since $C_2 \times C_{12}$ is abelian M/F is Galois.

(a) 3. Then, the number of subextensions satisfying $[K:M] = 6$ is equal to the number of subgroups of size 6. There are 3 such subgroups $\langle(1,2)\rangle, \langle(0,2)\rangle, \langle(1,4)\rangle$ and thus, 3 such subextensions.

(b) 3. The number of extensions satisfying $[M:F] = 6$ is equal to the number of subgroups of index 6 (order 4). These are also 3 in number, namely, $\langle(0,3)\rangle, \langle(1,3)\rangle, \langle(1,0),(0,6)\rangle$

(c) 3. The number of subextensions satisfying $\mathsf{Gal}(K/M) \cong C_6$ is equal to the number of cyclic subgroups of order 6 (since $C_2 \times C_3 \cong C_6$), which is 3 (part (a)).

(d) 3. The number of subextensions satisfying $\mathsf{Gal}(K/M) \cong C_6$ is equal to the number of cyclic subgroups with index 6 (order 4) (also recall, $C_2 \times C_3 \cong C_6$), which is 3 (part (b)).